

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    /* Create pipe */
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(EXIT_FAILURE);
    }

    /* Create first child */
    pid1 = fork();

    if (pid1 < 0)
    {
        perror("fork");
        exit(EXIT_FAILURE);
    }

    if (pid1 == 0)
    {
        /* First child: execute ls -l */

        close(pipefd[0]); // Close unused read end

        /* Redirect stdout to pipe */
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", (char *)NULL);

        perror("execlp ls");
        exit(EXIT_FAILURE);
    }

    /* Create second child */
    pid2 = fork();

    if (pid2 < 0)
    {
        perror("fork");
        exit(EXIT_FAILURE);
    }
}
```

```

if (pid2 == 0)
{
    /* Second child: execute grep ".c" */

    close(pipefd[1]); // Close unused write end

    /* Redirect stdin from pipe */
    dup2(pipefd[0], STDIN_FILENO);

    close(pipefd[0]);

    execlp("grep", "grep", ".c", (char *)NULL);

    perror("execlp grep");
    exit(EXIT_FAILURE);
}

/* Parent process */
close(pipefd[0]);
close(pipefd[1]);

/* Wait for both children */
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

return 0;
}

```

Screen Shots:

The screenshot shows a terminal window with the following commands and output:

```

sri_charan@SRICHARAN: ~/Pr
sri_charan@SRICHARAN:~$ ls
OSLab OSLab2 Practical 'Practical 1' Skill hello.c
sri_charan@SRICHARAN:~$ cd Practical
sri_charan@SRICHARAN:~/Practical$ gcc Practical5.c -o Practical5
sri_charan@SRICHARAN:~/Practical$ ./Practical5
-rw-r--r-- 1 sri_charan sri_charan 243 Aug 17 05:37 Permission.c
-rw-r--r-- 1 sri_charan sri_charan 684 Aug 3 05:43 Practical1.c
-rw-r--r-- 1 sri_charan sri_charan 1120 Aug 8 09:07 Practical2.c
-rw-r--r-- 1 sri_charan sri_charan 941 Aug 8 09:05 Practical3.c
-rw-r--r-- 1 sri_charan sri_charan 594 Aug 3 05:43 Practical4.c
-rw-r--r-- 1 sri_charan sri_charan 1383 Sep 15 16:10 Practical5.c
-rw-r--r-- 1 sri_charan sri_charan 1598 Sep 15 16:11 Practical6.c
-rw-r--r-- 1 sri_charan sri_charan 1315 Sep 15 16:11 Practical6_client.c
-rw-r--r-- 1 sri_charan sri_charan 1598 Sep 15 16:12 Practical6.c
sri_charan@SRICHARAN:~/Practical$ ls -l | grep ".c"
-rw-r--r-- 1 sri_charan sri_charan 243 Aug 17 05:37 Permission.c
-rw-r--r-- 1 sri_charan sri_charan 684 Aug 3 05:43 Practical1.c
-rw-r--r-- 1 sri_charan sri_charan 1120 Aug 8 09:07 Practical2.c
-rw-r--r-- 1 sri_charan sri_charan 941 Aug 8 09:05 Practical3.c
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-rw-r--r-- 1 sri_charan sri_charan 1315 Sep 15 16:11 Practical6_client.c
-rw-r--r-- 1 sri_charan sri_charan 1598 Sep 15 16:12 Practical6.c
sri_charan@SRICHARAN:~/Practical$

```

